Object-aware Contrastive Learning for Debiased Scene Representation

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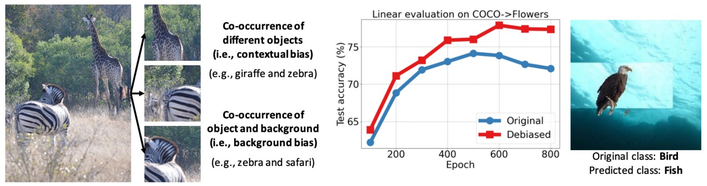
Code: https://github.com/alinlab/object-aware-contrastive

TL;DR. We propose a novel contrastive learning framework that tackles scene bias via object-aware augmentations.

Introduction

Self-supervised learning of visual representations is a fundamental problem of machine learning and computer vision. Contrastive learning (or positive-only methods) is a dominant approach for visual self-supervised learning.

Problem. Current contrastive learning suffer from the scene bias issue due to the co-occurrence of different objects, and spurious correlations between objects and background. The scene bias harms generalization of learned representations.



(a) Scene bias in random crop

(b) Performance drop

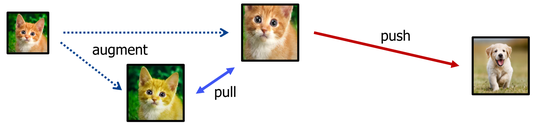
(c) Biased prediction

Contribution. We propose (a) self-supervised way of localizing objects, contrastive class activation map (ContraCAM), and (b) two object-aware data augmentations: object-aware random crop OA-Crop) and background mixup (BG-Mixup).

Contrastive Learning

Contrastive learning aims to learn an encoder $f(\cdot)$ that extracts a useful representation from unlabeled images. The learning is based on pulling similar (positive) samples and pushing dissimilar (negative) samples to each other.

- Positive samples come from the same sample but different augmentations.
- Negative samples are different samples (unnecessary for positive-only methods).

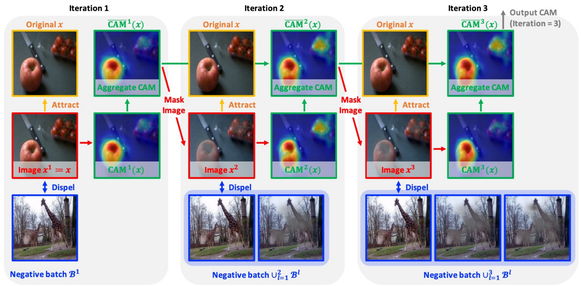


However, naively applying augmentations (e.g., random crop) to create positive samples make $f(\cdot)$ suffer from the scene bias.

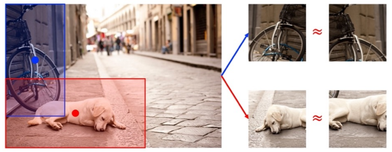
- We mitigate the problem via object-aware augmentations.

Object-aware Contrastive Learning

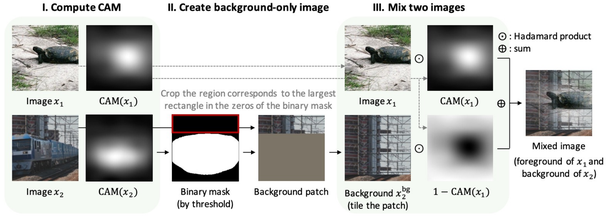
ContraCAM. To localize objects in an image, we extend the class activation map (CAM) for contrastive learning. Specifically, we find the most discriminative regions compared to the negative batch of images.



Object-aware random crop. To reduce contextual bias (from the co-occurrence of different objects), we restrict random crop around a single object.



Background mixup. To reduce background bias (from the spurious correlation of objects and background), we augment background-shifted samples.



Experiments

Localization. ContraCAM shows a competitive localization performance compared to the previous SOTA unsupervised localization method and supervised classifier CAM.



(a) CUB [46] (b) Flowers [26] (c) COCO [25] (d) ImageNet-9 [23]

Learning from multi-object images. OA-Crop improves representation learning from multi-object images, measured by linear evaluation and transfer learning.

Model	Network	OA-Crop	Test dataset						
			COCO-Crop	CIFAR10	CIFAR100	CUB	Flowers	Food	Pets
MoCov2	ResNet-50	-	74.30	77.58	53.26	22.90	72.09	59.70	59.25
MoCov2	ResNet-50	CAM	76.37 (+2.07)	84.10 (+6.52)	62.72 (+9.46)	25.46 (+2.56)	77.33 (+5.24)	62.01 (+2.31)	60.97 (+1.72)
MoCov2	ResNet-50	GT	76.44 (+2.14)	84.03 (+6.45)	62.81 (+9.55)	22.59 (-0.31)	75.09 (+3.00)	57.47 (-2.23)	57.67 (-1.58)
BYOL	ResNet-50	-	73.36	76.62	51.79	21.95	73.77	59.49	60.72
BYOL	ResNet-50	CAM	74.92 (+1.56)	82.79 (+6.17)	61.13 (+9.34)	24.34 (+2.39)	77.83 (+4.06)	61.83 (+2.34)	61.27 (+0.55)
BYOL	ResNet-50	GT	80.69 (+7.33)	85.92 (+9.30)	65.06 (+13.27)	28.68 (+6.73)	77.95 (+4.18)	64.63 (+5.14)	65.69 (+4.97)

Generalization on distribution shifts. BG-Mixup improves the generalizability of representation on background and distribution shifts (e.g., shape bias, corruption).

Model	Augmentation	Test dataset			
		ImageNet-Sketch-9	Stylized-ImageNet-9	ImageNet-R-9	ImageNet-C-9
MoCov2	Baseline	46.70±0.67	25.66±0.54	37.51±0.80	31.82±0.40
MoCov2	+Mixup [41]	51.18±0.88 (+4.48)	32.36±0.12 (+6.70)	41.00±0.12 (+3.49)	40.15±2.07 (+8.33)
MoCov2	+CutMix [42]	45.92±0.88 (-0.78)	26.46±0.68 (+0.80)	37.07±0.31 (-0.44)	32.29±0.60 (+0.47)
MoCov2	+BG-Mixup (ours)	**52.15**±0.93 (+5.45)	**33.36**±0.61 (+7.70)	**41.50**±0.45 (+3.99)	**44.39**±0.89 (+12.57)

Dataset	MoCov2			BYOL		
	Baseline	BG-Mixup (CAM)	BG-HardMix (GT)	Baseline	BG-Mixup (CAM)	BG-HardMix (GT)
Original ↑	89.17±0.49	90.73±0.05 (+1.56)	89.69±0.14 (+0.52)	87.30±0.61	89.30±0.02 (+2.00)	90.95±0.33 (+3.65)
Only-BG-B ↓	31.29±2.46	29.60±0.89 (-1.69)	26.44±1.63 (-4.85)	25.59±0.78	25.70±3.46 (+0.11)	27.28±0.04 (+1.69)
Only-BG-T ↓	44.91±0.16	41.95±0.38 (-2.96)	40.11±0.58 (-4.80)	42.83±0.51	39.94±0.52 (-2.89)	41.16±0.17 (-1.67)
Only-FG ↑	63.62±4.71	70.55±1.71 (+6.93)	72.68±0.69 (+9.06)	61.04±0.94	67.53±0.30 (+6.49)	72.63±1.13 (+11.59)
Mixed-Same ↑	80.98±0.34	84.13±0.33 (+3.15)	84.48±0.17 (+3.50)	79.30±0.31	81.28±0.53 (+1.98)	84.94±0.47 (+5.64)
Mixed-Rand ↑	60.34±0.66	66.89±0.54 (+6.55)	71.95±0.54 (+11.61)	58.03±0.85	63.83±0.53 (+5.80)	70.51±0.33 (+12.48)
Mixed-Next ↑	55.50±0.71	63.64±0.41 (+8.14)	70.25±0.14 (+14.75)	53.35±0.36	63.05±3.54 (+9.70)	66.81±0.08 (+13.46)
BG-Gap ↓	20.64±0.36	17.24±0.31 (-3.40)	12.53±0.69 (-8.11)	21.27±0.64	17.45±0.15 (-3.82)	14.44±0.56 (-6.83)